

Design Calculation Sheet — VBF-T3R1-CALC-001

Sheet: Ratings

Item | Value | Formula | Source

Q_nom | 3.08 | design nominal (fixed-point from measured 1C) | p_v12.json

I_1C | 3.08 | Q_nom x 1 | p_v12.json

I_5C | 15.4 | Q_nom x 5 | p_v12.json

I_4C | 12.32 | Q_nom x 4 | p_v12.json

Sheet: Performance

Metric | Value | Threshold (entry 0) | Verdict | Source

capacity_ah (1C) | 3.075 | 2 | PASS | r4_v12_1c.json

retention_5c | 0.97532 | 0.95 | PASS | r4_v12_derived.json

capacity_ah (5C) | 2.9991 | - | - | r4_v12_5c.json

T_max_K (4C@45C) | 329.7735 | 333.15 | PASS | r4_v12_derived.json

T_max_C | 56.62 | 60 | PASS | mechanical: K-273.15

anode_min_v (4C) | 0.01213 | > 0 (plated=false) | PASS | r4_v12_4c.json

power_density_w_kg | 32629.6 | 4000 | PASS | r4_v12_energy.json

Sheet: EnergyAndPower

Item | Value | Formula / note | Source

energy_wh | 11.09128009108955 | integral V(t) x I_1C dt | calc-energy

mass_kg | 0.029262302784 | sum layer thickness x (1-eps) x density x area | calc-energy

energy_density_wh_kg | 379.0296400443927 | energy/mass | calc-energy

energy_density_wh_l | 818.1582198142241 | energy/volume (stack only) | calc-energy

dcr_ohm | 0.004412872641653557 | (V[0] - V@10% time)/I_1C | calc-energy

power_density_w_kg | 32629.61467511318 | V_OC^2/(4 x DCR)/mass | calc-energy

midpoint_voltage_v | 3.843608531339204 | V at discharge time midpoint | calc-energy

electrolyte_included | False | contract excludes electrolyte mass | calc-energy

Sheet: Mass

Layer | Thickness [um] | Porosity | Density [kg/m3] | kg/m2 | Area [m2] | Mass [g] | Source

Positive electrode (NMC811) | 40 | 0.42 | 3699 | 0.075678400000000002 | 0.1027 | 7.772 | Chen2020 set

Negative electrode (graphite)	52	0.35	2060	0.0560066	0.1027	5.752	Chen2020 set
Separator (polyolefin)	12	0.47	1548	0.00252492	0.1027	0.259	Chen2020 set
Positive current collector (Al)	16	-	2702	0.0432	0.1027	4.437	Chen2020 set
Negative current collector (Cu)	12	-	8933	0.10752	0.1027	11.042	Chen2020 set
TOTAL	132		0.1027	29.26	calc-energy mass_kg		